Vaibhav Kotla

Email:kotlavaibhav@gmail.com **Tel:** +1 (919)-720-3324

https://www.linkedin.com/pub/vaibhav-kotla/91/571/ba6

https://github.com/kotlavaibhav

OBJECTIVE

To secure a software developer position where I can efficiently contribute my skills and abilities to the growth of the organization and build my professional career.

WORK EXPERIENCE

Software Development Intern **(June 2015 - August 2015)**

Seagate Technology, San Jose, CA

- Developed a multi-threaded software adapter in C++ and JAVA in Linux environment with TCP/IP protocols that helped to communicate with an NVME controller test software application on Windows. Reduced response time from minutes to under 10 seconds.
- Developed test scripts in python which reduced test time by several hundreds of hours annually. Prepared models for predicting performance for future testing.
- Debugged test software in a Linux/C++ environment and optimized the software to increase testing efficiency and speed by three hundred percent.

TECHNICAL SKILLS

- C, C++, Java/JEE, Python, Socket Programming, Operating Systems.
- HTML, CSS, AJAX, GitHub, LAMP, PHP, JavaScript.
- TCP/IP, BGP, L2, L3, UDP, HTTP, DHCP, IPv4, IPv6.
- Unix/Linux- Fedora, Macintosh, Ubuntu, Windows, VMWare virtualization, Hyper-V.
- Eclipse, NetBeans, Visual Basic 2015, Microsoft office, Wireshark, OPNET, Tcpdump, Putty.
- Excellent communication and presentation skills, an appetite for learning new technologies.

EDUCATION

M.S. in Computer Engineering, Specialization in Computer Networks **May 2016**

San Jose State University, San Jose, CA

Bachelor's in Electrical & Electronics Engineering **May 2013**

JNT University, Hyderabad

Relevant Courses

Object oriented programming, Computer Network Design, Network Programming & Applications, Network Security, Operating Systems, Computer Architecture, System Software, Mobile software system design, Business Intelligence Technologies, Network Architecture & protocols

ACADEMIC PROJECTS

Algorithm for authenticating and retrieving a user's password (C++)

- Developed a hash table search algorithm to encrypt a user's password and store it in a database.
- Resolved collisions using open addressing double hashing and maintained the load factor below 0.5 for extremely fast hashing.
- Retrieves stored password in O(1) time and decrypts the password to compare and to get the user logged in to the software.

Database Implementation (C++)

- Implemented a Heap file API and a multithreaded 2 Phase Multi-way merge sort for sorting of a database records into a sorted file.
- Implemented some relational operators and aggregate functions like Select, Project, Join, Group By, SUM.
- Estimated the costs of various join orders to choose the optimal one.

Multi-threaded DHCP Server (Java)

- Developed a multi-threaded DHCP server in Java with an IP address pool and database to service incoming client connections from the shared IP address pool.
- Systems will send its MAC address and in return receive an IP address. The server maintains a list of IP addresses allocated.

Multi-threaded Bank Server (JAVA)

- Multi-threaded bank server with multi-process clients using synchronization to eliminate race conditions.
- To prevent simultaneous access, database access is protected with a mutex lock.

Webserver Design with Photo Sharing (PHP, HTML)

- Developed a web server using PHP, JavaScript & HTML web design using bootstrap, passwords using BCRYPT algorithms.
- Once any file is uploaded or picture taken with mobile camera is uploaded the image is displayed in the gallery depending on the user criteria (public or private). All the images and videos will be uploaded to the peer servers using CURL.

Dictionary Implementation (C++)

- Developed a binary search tree which can store key, value pairs in it.
- Each node has a comparable key (and an associated value). Values can be inserted, deleted and retrieved.

Network Security (Java)

- Built an application on Java which enables simple secure data transfer on a server-client model, after establishing an SSL like connection.
- For secure, authenticated and integrity protected data transfer, built routines for algorithms including 3DES, and RSA.

Virtual File System Implementation (LINUX)

- Implemented and initialized a VFS with superblock setup and created root directories and files.
- Studied the architecture of VFS, relationship between objects, interaction between processes and VFS objects in the common file model.

Client Server protocol for improving routing mechanism (JAVA)

- Developed a client server protocol implementation of a gateway protocol for improving routing mechanism and updating routing tables on different connected gateways.

Network Simulation (OPNET)

- Performed network simulation and modeling using OPNET. Developed models of WLAN, VLAN and Network Design with different users, hosts, and services.
- Configured network traffic parameters to increase efficiency by 40% and predicted future performance for these models.

Mobile App (Phone Gap) https://github.com/kotlavaibhav/mobileapplication

- Developed a grading application where professors can grade students and calculate the result. It includes customized settings to set the cut-off for each grades with sliders.
- The application sends SMS alerts to students and shows individual student information, courses.